Owing to our ansatz, we would expect that the computed average of some observable has behavior like

$$\langle \vartheta \rangle_h = \langle \vartheta \rangle + h^2 \langle \vartheta f_{2,0} \rangle + \mathcal{O}(\varepsilon h^2 + h^4).$$

At sufficiently small values of ε (and hence, large values of the friction coefficient γ) we would expect that the $\mathcal{O}(h^2)$ remainder will not be visible to us in our experiments, and we will be able to correct our average to $\mathcal{O}(h^4)$. The symmetric discretization schemes we have considered here will all be consistent in the overdamped ($\gamma \to \infty$) limit, and we would expect the correction function for the corresponding limiting schemes to be given by $f_{2,0}$.

7.9.3 *Superconvergence of the* ⟦BAOAB⟧ *Scheme*

In the regime of sufficiently large γ and small h, our doubly-asymptotic ansatz for the leading order correction to a function $\vartheta(q,p)$ is

$$\langle \vartheta \rangle = \langle \vartheta \rangle_h - h^2 \langle \vartheta f_{2,0} \rangle + O\left(\frac{h^2}{\gamma} + h^4\right), \tag{7.37}$$

where $f_{2,0}$ is method-dependent. The ⟦BAOAB⟧ scheme has leading order correction

$$f_{2,0,[\![\mathrm{BAOAB}]\!]}(\boldsymbol{q},\boldsymbol{p}) = \frac{1}{8}\left(\Delta U - \beta \boldsymbol{p}^T \nabla^2 U(\boldsymbol{q})\boldsymbol{p}\right),$$

which has the interesting property that its marginal average with respect to momentum is zero:

$$\bar{\pi} f_{2,0,[\![\mathrm{BAOAB}]\!]} = \int_{\mathcal{P}} f_{2,0,[\![\mathrm{BAOAB}]\!]}(\boldsymbol{q},\boldsymbol{p}) \rho_\beta(\boldsymbol{q},\boldsymbol{p}) \mathrm{d}\omega_p = 0.$$

This is a very favorable quality for configurational sampling, where we consider averages of functions solely of position, as we can split up the integral such that

$$\begin{aligned}
\langle \vartheta(\boldsymbol{q}) f_{2,0,[\![\mathrm{BAOAB}]\!]} \rangle &= \int_{\mathcal{D}} \vartheta(\boldsymbol{q}) f_{2,0,[\![\mathrm{BAOAB}]\!]} \rho_\beta \mathrm{d}\omega, \\
&= \int_{\mathcal{Q}} \vartheta(\boldsymbol{q}) \int_{\mathcal{P}} f_{2,0,[\![\mathrm{BAOAB}]\!]} \rho_\beta \mathrm{d}\omega_p \mathrm{d}\omega_q, \\
&= 0.
\end{aligned}$$